Semester Assignment (OOP Using Java Programming)

YouthEmpowermentProject (YEP)

YouthEmpowermentProject (YEP) – Youth Digital Empowerment and Virtual Exchange Platform

1. Background

Young people in Uganda and the wider East African region face growing challenges related to unemployment, digital skills gaps, misinformation, responsible use of Artificial Intelligence (AI), and limited opportunities for international collaboration.

The YEP is a proposed initiative designed to connect young people aged 18–35 from Uganda, and other East African countries through an online platform. The platform will provide opportunities for digital learning, Virtual Exchange activities, teamwork, mentorship, and development of simple youth-related Impact Projects.

The platform will support two levels of participation. Level One Virtual Exchanges will involve large groups of young people participating in online learning and discussion sessions focused on digital literacy, AI, critical thinking, media literacy, and intercultural communication.

Level Two Virtual Exchanges will allow selected participants to form teams and develop practical solutions to challenges in areas such as employment, education, agriculture, health, climate change, and digital inclusion.

In this semester project, students will analyse the YEP case and propose a simple suitable web-based information system that supports youth registration, learning activities, team formation, project development, assessment, communication, and recognition of achievements.

2. Assignment Requirements

Working in groups, students must:

- Conduct detailed research on youth digital skills, employability, AI literacy, and Virtual Exchange opportunities in Uganda and East Africa.
- Use at least five (5) credible sources, such as government reports, academic papers, development-agency reports, reputable news sources, or recognised technology organisations.
- Analyse the needs of different YEP stakeholders, including youth participants, facilitators, mentors, universities, employers, partner organisations, and administrators.
- Identify the functional and non-functional requirements of the proposed YEP system.
- Design a suitable web-based solution for managing YEP activities.
- Apply appropriate principles of Object-Oriented Programming and other techniques that will seem helpful such as software engineering principles, web application development, database design depending on the course requirements.

3. Assignment Objectives

Students are expected to:

1. Research the digital skills and youth empowerment challenges affecting Uganda and East Africa.
2. Analyse the requirements and challenges of the YEP stakeholders.
3. Design a suitable system and may be a web app also for managing youth participation and Virtual Exchange activities.
4. Apply appropriate system analysis and design techniques.
5. Develop a suitable UML class diagram showing the major system classes and their relationships.
6. Design an appropriate database structure for managing users, teams, activities, projects, assessments, and certificates.
7. Provide pseudo-code for important system classes, methods, or processes.
8. Consider important issues such as security, privacy, accessibility, mobile-first design, low-bandwidth access, and responsible use of AI.
9. Develop a simple prototype demonstrating selected YEP system functions.

4. Expected Output / System Expectations

Students should produce the following:

1. Written Research and System Analysis Report – PDF
2. Stakeholder Analysis – PDF/document
3. Functional and Non-Functional Requirements – PDF/document
4. UML Use Case Diagram – image/PDF
5. UML Class Diagram – image/PDF
6. Entity-Relationship (ER) Diagram – image/PDF
7. Data Flow Diagram (DFD) – image/PDF
8. Pseudo-code for Key System Classes and Methods – text/PDF
9. User Interface/Wireframe Designs for major YEP pages – image/PDF
10. Simple Prototype of the YEP system – bonus/where applicable

Good Luck